
# OPERATING SYSTEMS
# UE24CS242B
## Process Management

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## Slides Credits for all the PPTs of this course

- The slides/diagrams in this course are an **adaptation**, **combination**, and **enhancement** of material from the following resources and persons:

1. Slides of Operating System Concepts, Abraham Silberschatz, Peter Baer Galvin, Greg Gagne - 9th edition 2013 and some slides from 10th edition 2018
2. Some conceptual text and diagram from Operating Systems - Internals and Design Principles, William Stallings, 9th edition 2018
3. Some presentation transcripts from A. Frank – P. Weisberg
4. Some conceptual text from Operating

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# OPERATING SYSTEMS

## Process Concept

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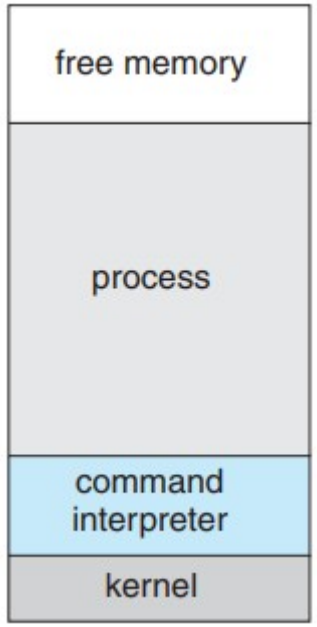
# OPERATING SYSTEMS
## Process Concept

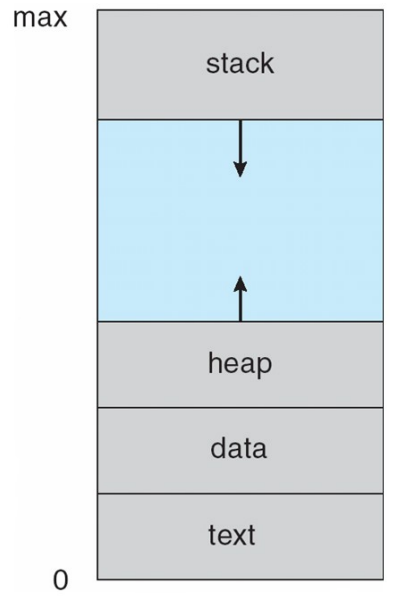

- An operating system executes a variety of programs:
  - Batch system – **jobs**
  - Time-shared systems – **user programs** or **tasks**
- Textbook uses the terms ***job*** and ***process*** almost interchangeably
- **Process** – a program in execution; process execution must progress in sequential fashion
- Program is ***passive*** entity stored on disk (**executable file**), process is ***active***
  - Program becomes process when executable file loaded into memory
- Execution of program started via GUI mouse clicks, command line entry of its name, etc
- One program can be several processes



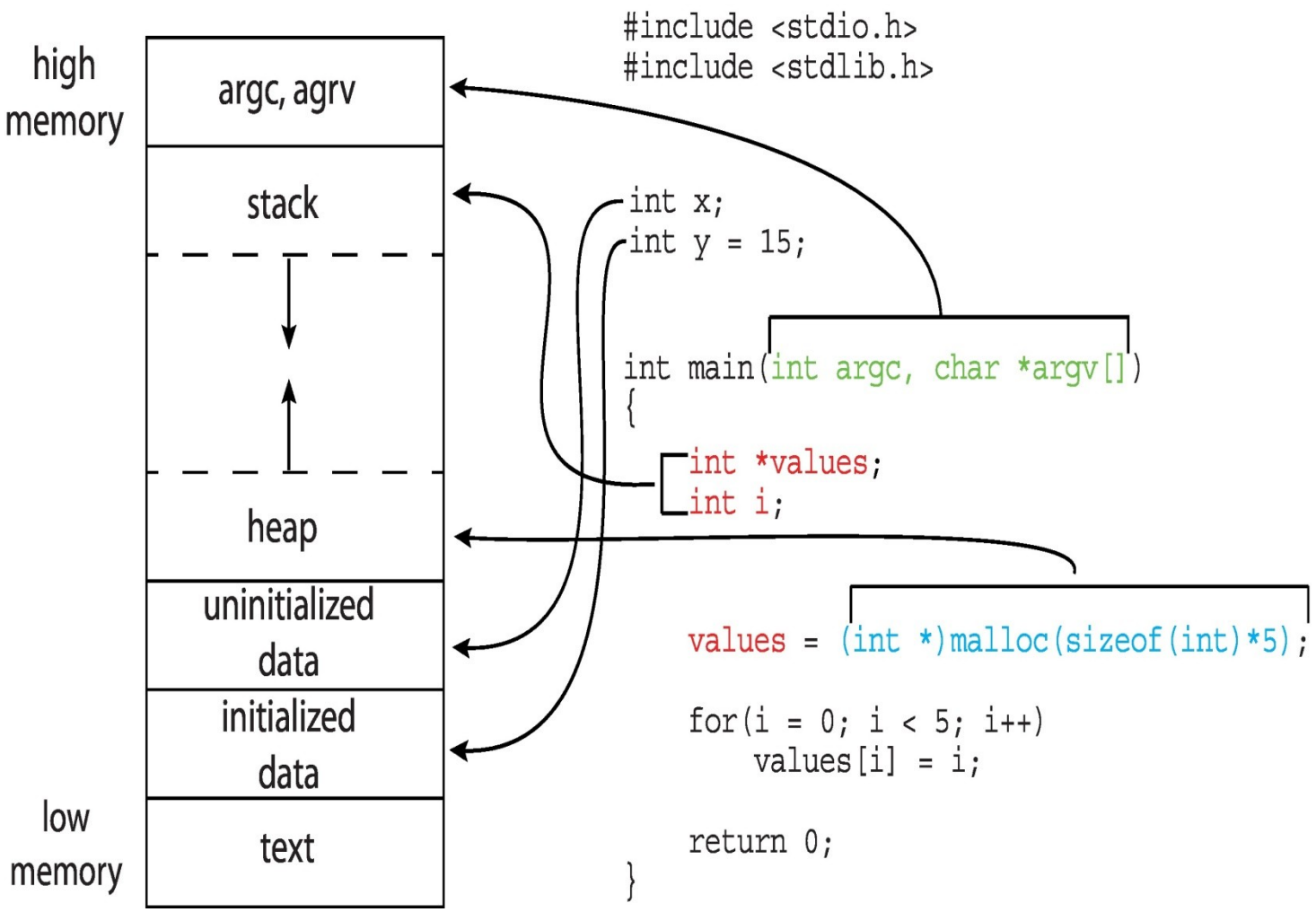
# Process Concept

## Structure of a process in memory

- The program code, also called **text section.**
  - Includes current activity including **program counter**, processor registers
- **Stack** containing temporary data
  - Function parameters, return addresses, local variables
- **Data section** containing global variables
- **Heap** containing memory dynamically allocated



# Memory Layout of a C Program



# OPERATING SYSTEMS

## Memory Layout of a process

```
suresh@PES1UG19CS553:~/OS$ gcc -o q1b.exe q1b.c
suresh@PES1UG19CS553:~/OS$ ./q1b.exe 5
Process id: 23059
```

```
suresh@PES1UG19CS553:~$ pmap 23059
23059:   ./q1b.exe 5
000055cefc914000      4K r-x-- q1b.exe
000055cefcb14000      4K r---- q1b.exe
000055cefcb15000      4K rw--- q1b.exe
000055cefe6e7000    132K rw---   [ anon ]
00007f26ef253000   1948K r-x-- libc-2.27.so
00007f26ef43a000   2048K ----- libc-2.27.so
00007f26ef63a000     16K r---- libc-2.27.so
00007f26ef63e000      8K rw--- libc-2.27.so
00007f26ef640000     16K rw---   [ anon ]
00007f26ef644000    164K r-x-- ld-2.27.so
00007f26ef857000      8K rw---   [ anon ]
00007f26ef86d000      4K r---- ld-2.27.so
00007f26ef86e000      4K rw--- ld-2.27.so
00007f26ef86f000      4K rw---   [ anon ]
00007ffda2b15000    136K rw---   [ stack ]
00007ffda2bae000     12K r----   [ anon ]
00007ffda2bb1000      8K r-x--   [ anon ]
ffffffffff600000      4K --x--   [ anon ]
 total             4524K
```

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# Memory Layout of a process

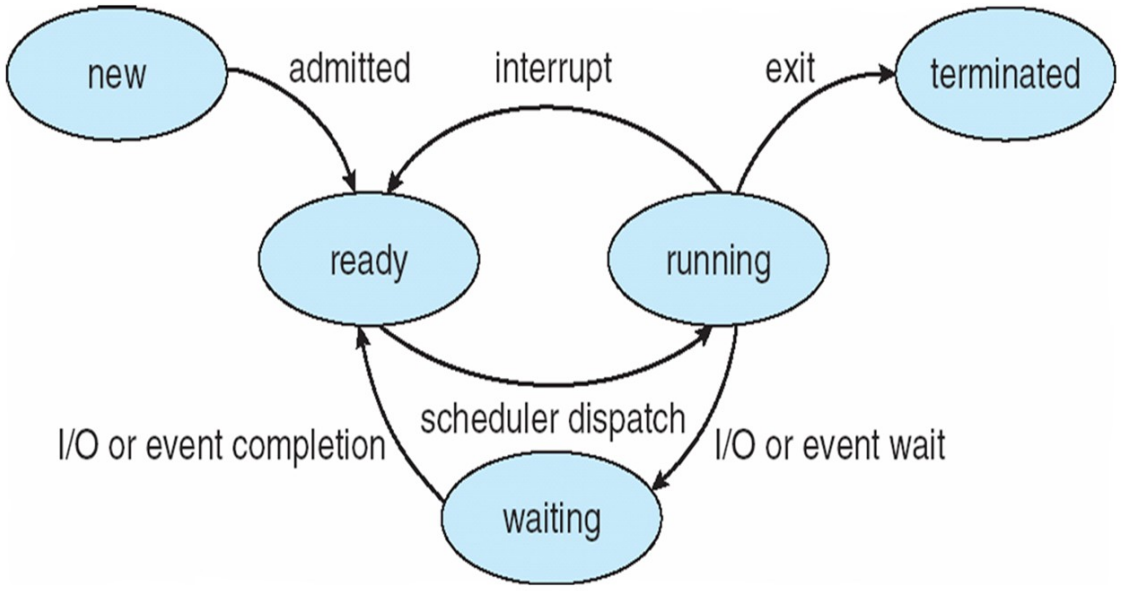


```
suresh@PES1UG19CS553:~$ cat /proc/23059/maps
55cefc914000-55cefc915000 r-xp 00000000 08:07 4338399                    /home/suresh/OS/q1b.exe
55cefcb14000-55cefcb15000 r--p 00000000 08:07 4338399                    /home/suresh/OS/q1b.exe
55cefcb15000-55cefcb16000 rw-p 00001000 08:07 4338399                    /home/suresh/OS/q1b.exe
55cefe6e7000-55cefe708000 rw-p 00000000 00:00 0                          [heap]
7f26ef253000-7f26ef43a000 r-xp 00000000 08:07 526643                     /lib/x86_64-linux-gnu/libc-2.27.so
7f26ef43a000-7f26ef63a000 ---p 001e7000 08:07 526643                     /lib/x86_64-linux-gnu/libc-2.27.so
7f26ef63a000-7f26ef63e000 r--p 001e7000 08:07 526643                     /lib/x86_64-linux-gnu/libc-2.27.so
7f26ef63e000-7f26ef640000 rw-p 001eb000 08:07 526643                     /lib/x86_64-linux-gnu/libc-2.27.so
7f26ef640000-7f26ef644000 rw-p 00000000 00:00 0
7f26ef644000-7f26ef66d000 r-xp 00000000 08:07 526515                     /lib/x86_64-linux-gnu/ld-2.27.so
7f26ef857000-7f26ef859000 rw-p 00000000 00:00 0
7f26ef86d000-7f26ef86e000 r--p 00029000 08:07 526515                     /lib/x86_64-linux-gnu/ld-2.27.so
7f26ef86e000-7f26ef86f000 rw-p 0002a000 08:07 526515                     /lib/x86_64-linux-gnu/ld-2.27.so
7f26ef86f000-7f26ef870000 rw-p 00000000 00:00 0
7ffda2b15000-7ffda2b37000 rw-p 00000000 00:00 0                          [stack]
7ffda2bae000-7ffda2bb1000 r--p 00000000 00:00 0                          [vvar]
7ffda2bb1000-7ffda2bb3000 r-xp 00000000 00:00 0                          [vdso]
ffffffffff600000-ffffffffff601000 --xp 00000000 00:00 0                  [vsyscall]
suresh@PES1UG19CS553:~$
```

# Process State

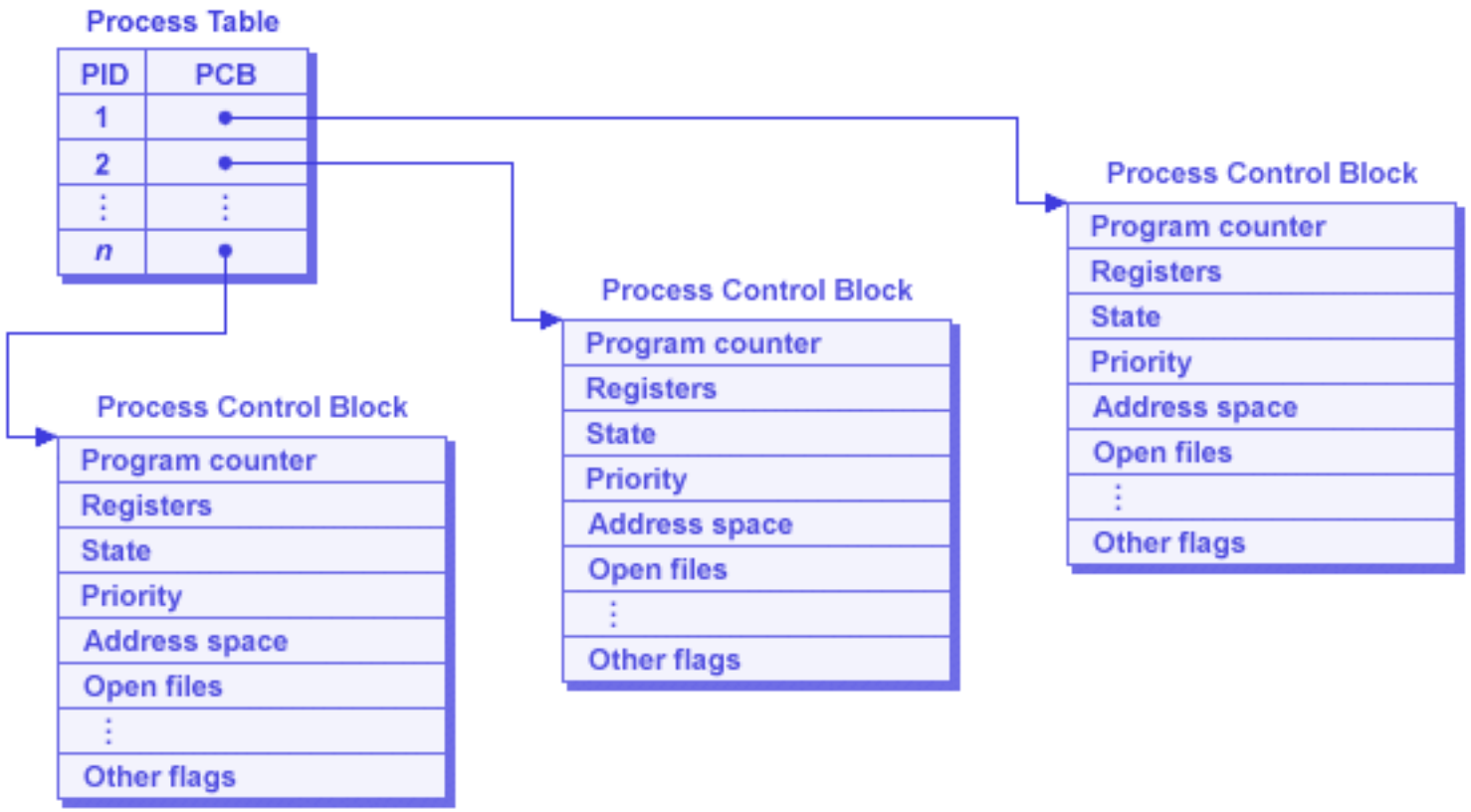
As a process executes, it changes **state**

- **New**: The process is being created
- **Running**: Instructions are being executed
- **Waiting**: The process is waiting for some event to occur
- **Ready**: The process is waiting to be assigned to a processor
- **Terminated**: The process has



# OPERATING SYSTEMS

## Process Control Block (PCB)

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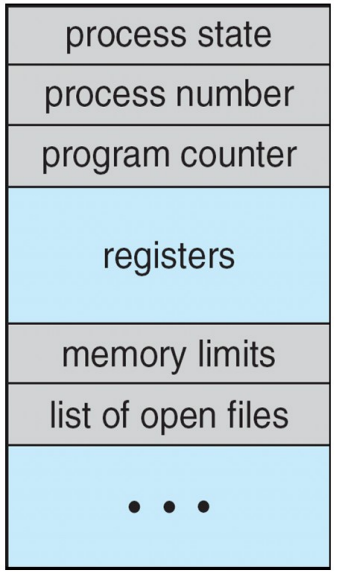
# OPERATING SYSTEMS

## Process Control Block (PCB)

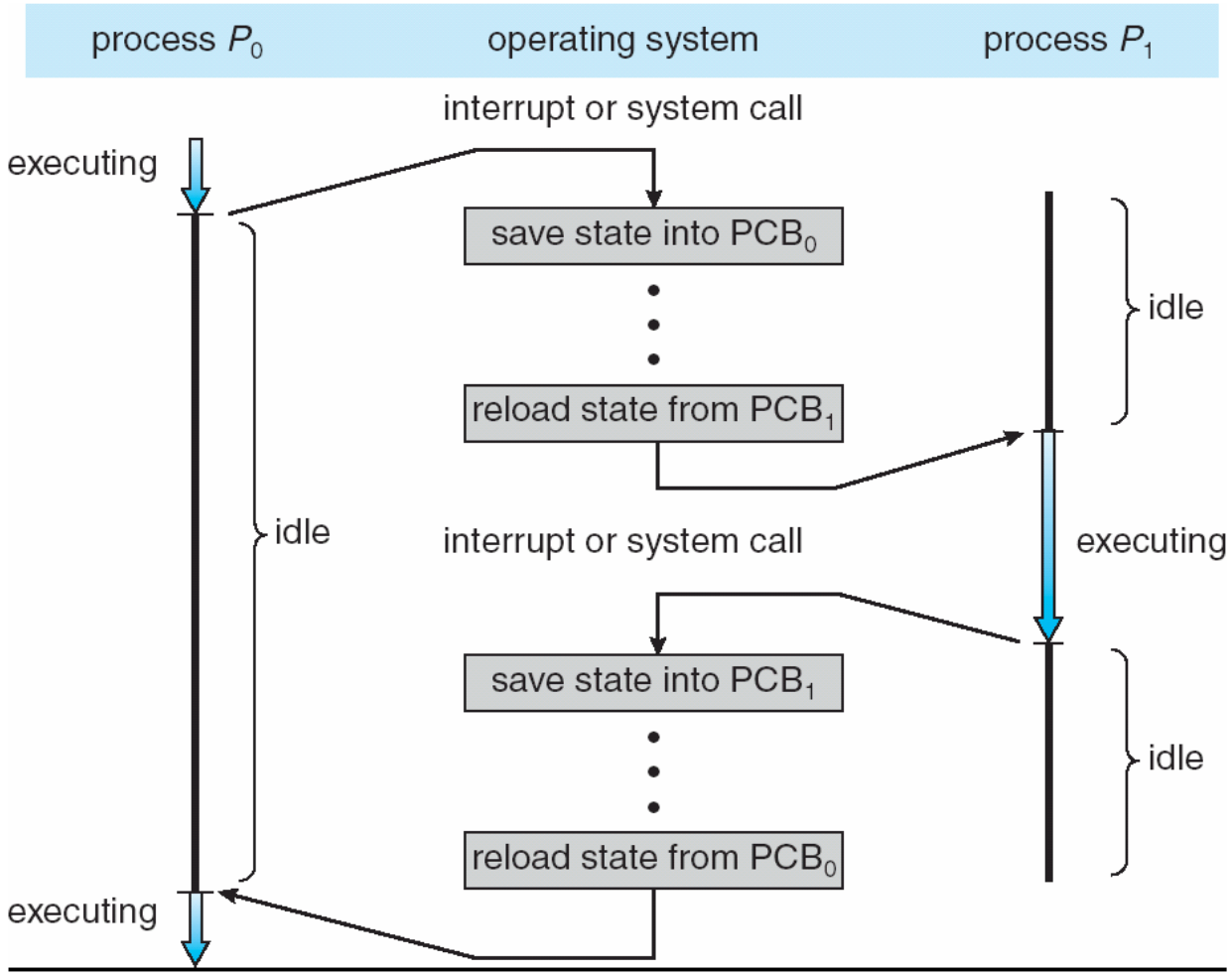

Each process is represented in the operating system by a Process Control Block (also called **task control block**)

- Process state – running, waiting, etc
- Program counter – location of instruction to next execute
- CPU registers – contents of all process-centric registers
- CPU scheduling information- priorities, scheduling queue pointers
- Memory-management information – memory allocated to the process
- Accounting information – CPU used, clock time elapsed since start, time limits
- I/O status information – I/O devices allocated to process, list of open files



# OPERATING SYSTEMS

## CPU switch from process to process

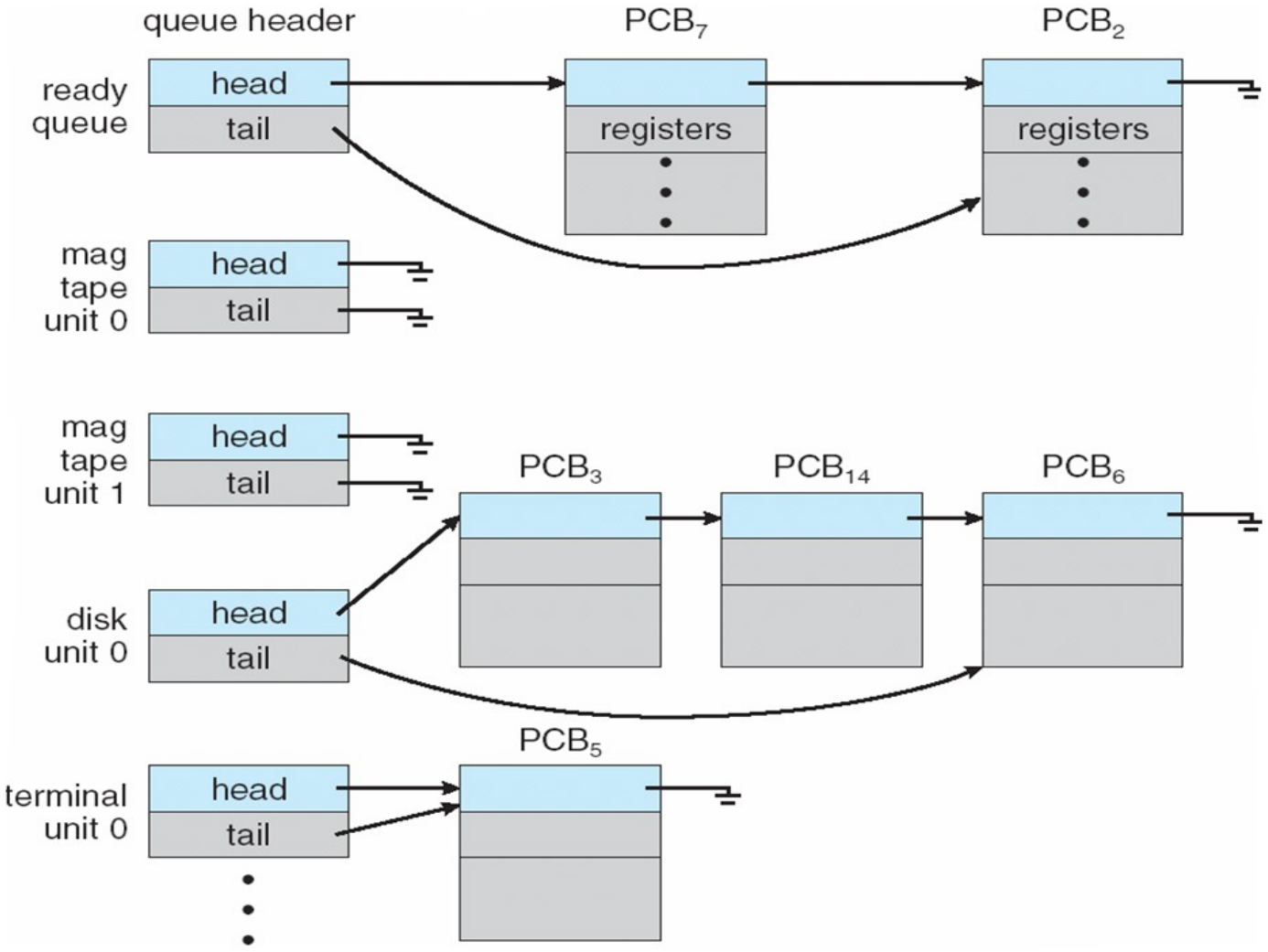


# Process Scheduling

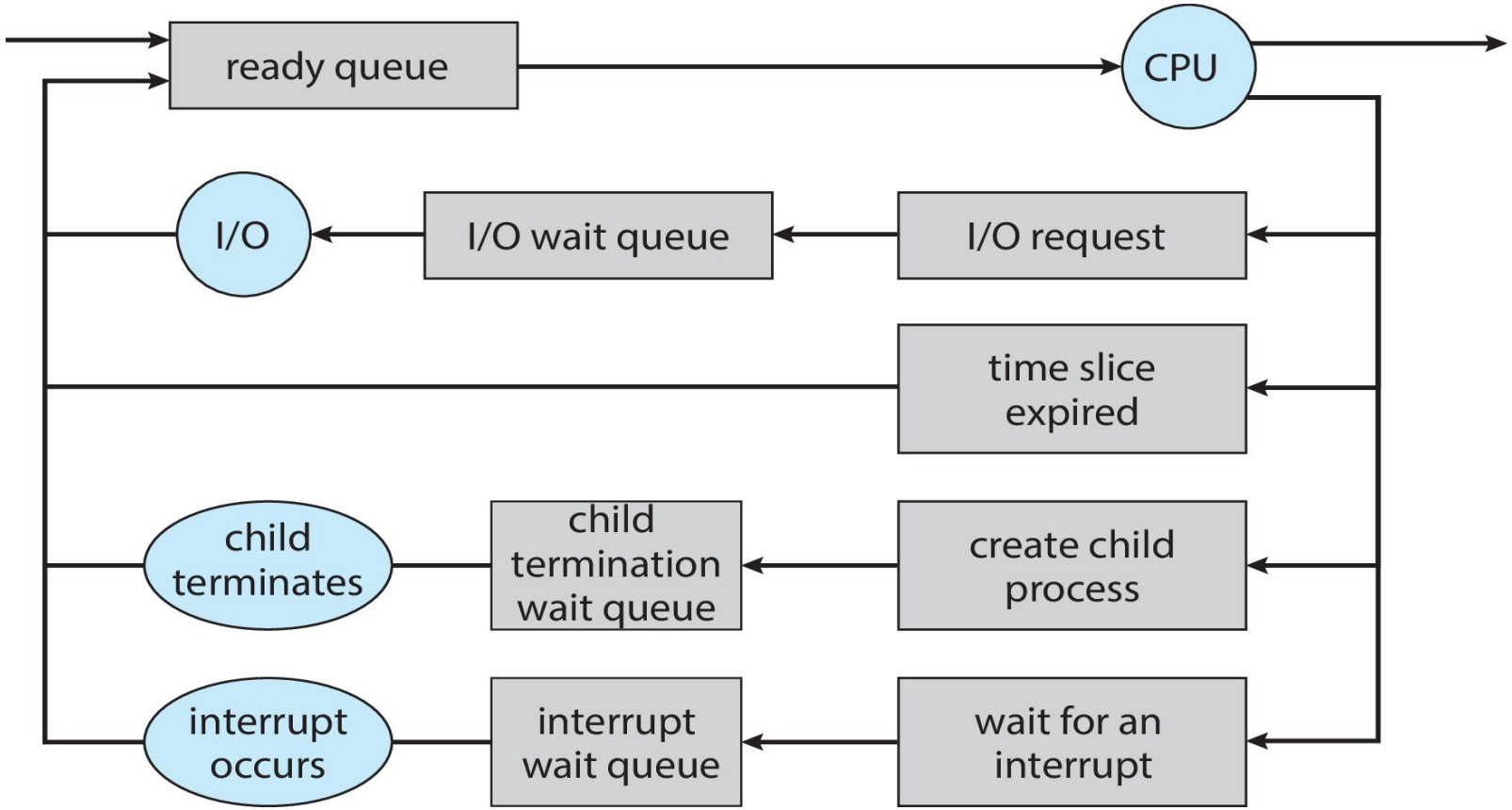
- Maximize CPU use, quickly switch processes onto CPU for time sharing
- **Process scheduler** selects among available processes for next execution on CPU
- Maintains **scheduling queues** of processes
  - **Job queue** – set of all processes in the system
  - **Ready queue** – set of all processes residing in main memory, ready and waiting to execute
  - **Device queues** – set of processes waiting for an I/O device
  - Processes migrate among the various queues

# OPERATING SYSTEMS

## Ready Queue And Various I/O Device Queues



# Representation of Process Scheduling



# Schedulers

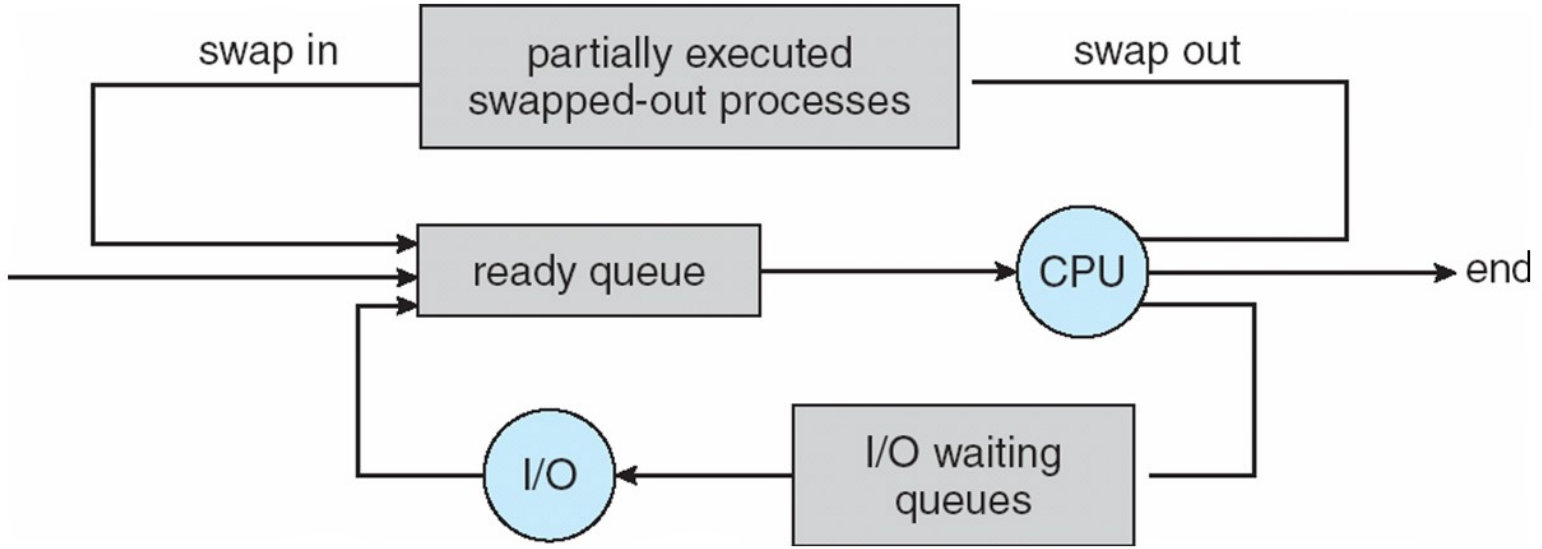
- **Short-term scheduler** (or **CPU scheduler**) – selects which process should be executed next and allocates CPU
  - Sometimes the only scheduler in a system
  - Short-term scheduler is invoked frequently (milliseconds) ⇒ (must be fast)
- **Long-term scheduler** (or **job scheduler**) – selects which processes should be brought into the ready queue
  - Long-term scheduler is invoked infrequently (seconds, minutes) ⇒ (may be slow)
  - The long-term scheduler controls the **degree of multiprogramming**

## Schedulers


- Processes can be described as either:
  - **I/O-bound process** – spends more time doing I/O than computations, many short CPU bursts
  - **CPU-bound process** – spends more time doing computations; few very long CPU bursts
- Long-term scheduler strives for good ***process mix***

# Representation of Process Scheduling

- **Medium-term scheduler** can be added if degree of multiple programming needs to decrease
  - Remove process from memory, store on disk, bring back in from disk to continue execution: **swapping**



# Context Switch

- When CPU switches to another process, the system must **save the state** of the old process and load the **saved state** for the new process via a **context switch**
- **Context** of a process represented in the PCB
- Context-switch time is overhead; the system does no useful work while switching
  - The more complex the OS and the PCB → the longer the context switch

# Operations on Processes

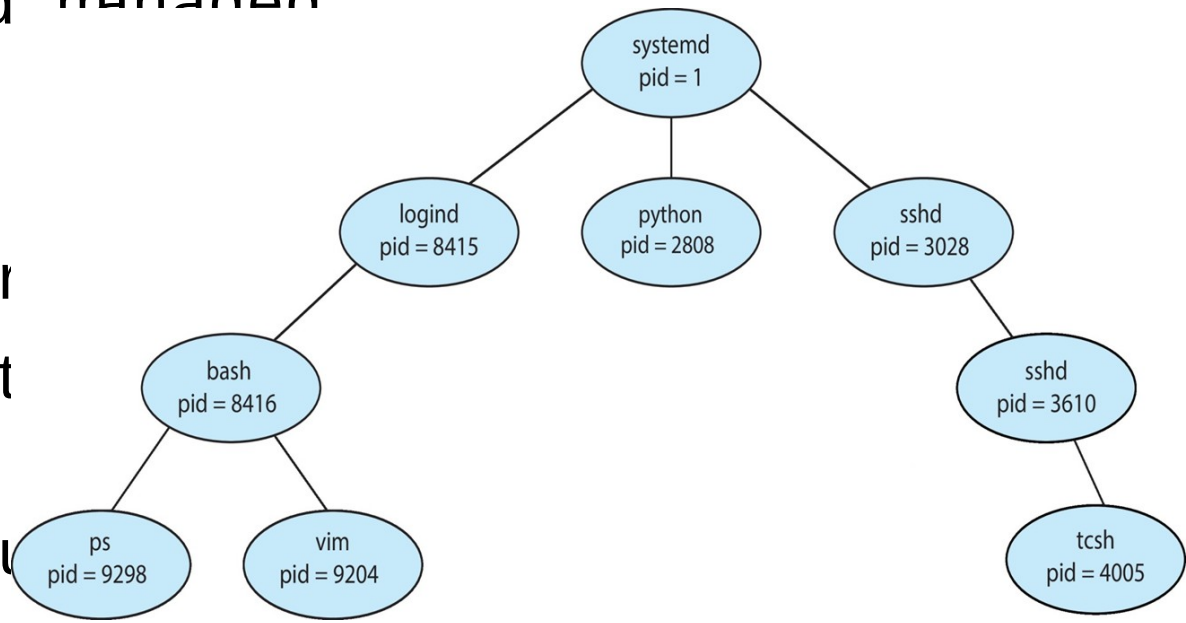
System must provide mechanisms for:

- process creation
- process termination

# OPERATING SYSTEMS

## Process Creation

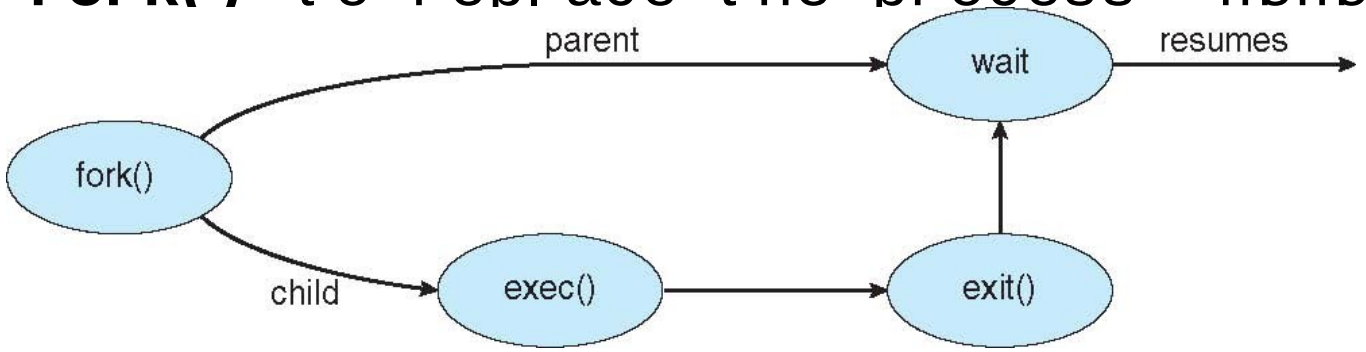
- **Parent** process creates **children** processes, which, in turn create other processes, forming a **tree** of processes
- Generally, process identified and managed via a **process identifier** (**pid**)
- Resource sharing options
  - Parent and children share all r
  - Children share subset of parent resources
  - Parent and child share no resou
- Execution options
  - Parent and children execute concurrently



# OPERATING SYSTEMS

## Process creation using fork()

- Address space
  - Child duplicate of parent
  - Child has a program loaded into it
- UNIX examples
  - **fork()** system call creates new process
  - **exec()** system call used after a **fork()** to replace the process' memory



# C Program forking Separate Process


```
#include <sys/types.h>
#include <stdio.h>
#include <unistd.h>

int main()
{
pid_t pid;

    /* fork a child process */
    pid = fork();

    if (pid < 0) { /* error occurred */
       fprintf(stderr, "Fork Failed");
       return 1;
    }
    else if (pid == 0) { /* child process */
       execlp("/bin/ls","ls",NULL);
    }
    else { /* parent process */
       /* parent will wait for the child to complete */
       wait(NULL);
       printf("Child Complete");
    }

    return 0;
}
```

# Process Termination

- Process executes last statement and then asks the operating system to delete it using the **exit()** system call.
  - Returns status data from child to parent (via **wait()**)
  - Process' resources are deallocated by operating system
- Parent may terminate the execution of children processes using the **abort()** system call. Some reasons for doing so:
  - Child has exceeded allocated resources
  - Task assigned to child is no longer required
  - The parent is exiting and the operating systems does not allow a child to continue if its parent terminates

# OPERATING SYSTEMS

## Process Termination


- Some operating systems do not allow child to exist if its parent has terminated. If a process terminates, then all its children must also be terminated.
  - **cascading termination.** All children, grandchildren, etc. are terminated.
  - The termination is initiated by the operating system
- The parent process may wait for termination of a child process by using the **wait()** system call. The call returns status information and the pid of the terminated process
  - **pid = wait(&status);**

# Process Termination


- What happens if parent process terminates before child process
- What happens if child process terminates before parent( i.e When the parent process in sleep)
- How to know the state of the process?


# THANK YOU

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